MZUZU UNIVERSITY

FACULTY OF SCIENCE, TECHNOLOGY AND INNOVATION

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY

ICT 1102: INTRODUCTION TO PROGRAMMING

END OF SEMESTER EXAMINATION

DATE: AUGUST, 2018 **TIME ALLOWED: 3 HOURS**

INSTRUCTIONS

1. This paper contains **FIVE** questions.
2. Answer **all** questions.
2. Marks for each question are indicated.
3. Write your answers clearly.
4. Indicate the question number for each answer.
5. This paper contains **TWO** pages. Please check.

Question One

a) Explain and give ***one*** example of each of the following language models:

   i) Imperative languages. **[3 marks]**

   ii) Functional languages. **[3 marks]**

b) Answer the following questions about structured programming.

   i) Explain *Structured Programming* as a programming methodology. **[3 marks]**

   ii) What are the ***three*** *main advantages* of structured programming over unstructured programming? **[3 marks]**

c) Answer the following questions concerning *language generations*.

   i) Name ***two*** *language generations* that allowed users to write programs that were *machine dependent*. **[2 marks]**

   ii) State ***two*** differences between the given language generations in ***c(i)*** above. **[2 marks]**

d) Write a pseudocode for the program that prints even numbers from 0 to 20 inclusive.

[5 marks]

Question Two

a) Write a data structure that depicts the data as follows: Information on customer orders is stored up in an order file. Each order consists of a header record holding the order number and the customer number, followed by a group of detail records. Each detail record carries the amount debited for a different product. We are required to produce a report showing the total amount debited for each order. **[7 marks]**

b) Write a pseudo code program to find the *arithmetic mean* of a list of *positive* numbers, and prints the arithmetic mean. The list of numbers is not known in advance. Terminate the input of numbers with *zero*. **[5 marks]**

c) Write a pseudo code for the program that accepts five exam grades from the user and computes the average of the grades. Output the average of the five exams. Together with the average, also include a statement "Well done!" in the output if the average is greater than or equal to 60, otherwise output "You need to pull up your socks!" **[6 marks]**

Question Three

a) Write a pseudo code for the program that prompts the user to enter a starting number between 1 and 100 and displays the entered number to the screen then calculates the sum of the numbers from the entered number to 200 and displays the result to the screen.

[10 marks]

b) Create a program that prints your name a hundred times. Do three versions of this program using a while loop, a do-while loop and a for-loop. **[12 marks]**

c) Mention any ***five*** key questions to ask when evaluating programming languages for use in a project. **[5 marks]**

d) Distinguish between the following terms:

i) Procedure and function. **[2 marks]**

ii) Global variable and local variable. **[2 marks]**

iii) Actual parameter and formal parameter. **[2 marks]**

END OF QUESTION PAPER